Role of the Vishniac Magnetic Helicity Flux in Mean-field Galactic Dynamos

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Gayathri K

Enrollment No. PHYS11202013019

under the guidance of

Dr. Luke Chamandy

Reader-F

to the

School of Physical Sciences

National Institute of Science Education and Research

Bhubaneswar

May, 2025

to my parents

DECLARATION

I hereby declare that I am the sole author of this thesis in partial fulfillment of the requirements for a postgraduate degree from National Institute of Science Education and Research (NISER). I authorize NISER to lend this thesis to other institutions or individuals for the purpose of scholarly research.

Signature of the Student
Date: 22nd May, 2025

The thesis work reported in the thesis entitled **Role of the Vishniac Magnetic Helicity Flux in Mean Field Galactic Dynamos** was carried out under my supervision, in the School of Physical Sciences at NISER, Bhubaneswar, India.

Signature of the thesis supervisor
School: School of Physical Sciences
Date: 22nd May, 2025

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ABSTRACT

Magnetic fields are crucial to the evolution of galaxies, but the mechanisms amplifying and sustaining large-scale fields remain unclear. In disc galaxies, the evolution of large-scale magnetic fields is explained through mean-field dynamo theory which focuses on the mean magnetic field ($\overline{\mathbf{B}}$), while the role of random magnetic fields ($\mathbf{b}$) is often overlooked. These smaller-scale fields serve as a helicity reservoir. Without effective fluxes to expel excess helicity, the buildup of magnetic helicity can suppress the α-effect. Helicity fluxes can counteract this by redistributing helicity. In this study, we incorporate one such helicity flux, the Vishniac flux, into numerical models for the first time, building on the analytical framework of Gopalakrishnan & Subramanian (2023). This thesis employs an independently developed numerical simulation code that solves a one-dimensional model in the vertical (z) direction, based on the thin-disc and axisymmetry approximations. A dynamical quenching framework is implemented, simultaneously solving the mean-field induction equation and the magnetic helicity evolution equation. We assess the impact of the Vishniac flux on the evolution of galactic magnetic fields in the presence of a finite diffusive flux and varying strengths of small-scale magnetic fields, parametrized by ξ_0, the ratio of small-scale magnetic to kinetic energy density. An analytical study of the model is also performed using the no-z approximation. Our results show that the Vishniac flux undergoes a sign reversal at a critical value $\xi_{\rm crit}$, marking a transition from dynamo suppression to enhancement. At high super-equipartition values $\xi_{\rm growth}$, the flux alone is capable of driving the dynamo even in the absence of kinetic helicity, by generating an effective α-effect through helicity transport. While such strong small-scale magnetic fields are not unphysical, they are also not typically realistic in the context of galactic discs. The results suggest the possibility of a magnetically driven dynamo mechanism under specific conditions, such as strong rotation and super-equipartition small-scale magnetic fields. This study highlights the potential role of helicity fluxes in sustaining large-scale magnetic fields and motivates further investigation into magnetic contributions to dynamo action.

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Chapter 1

Introduction

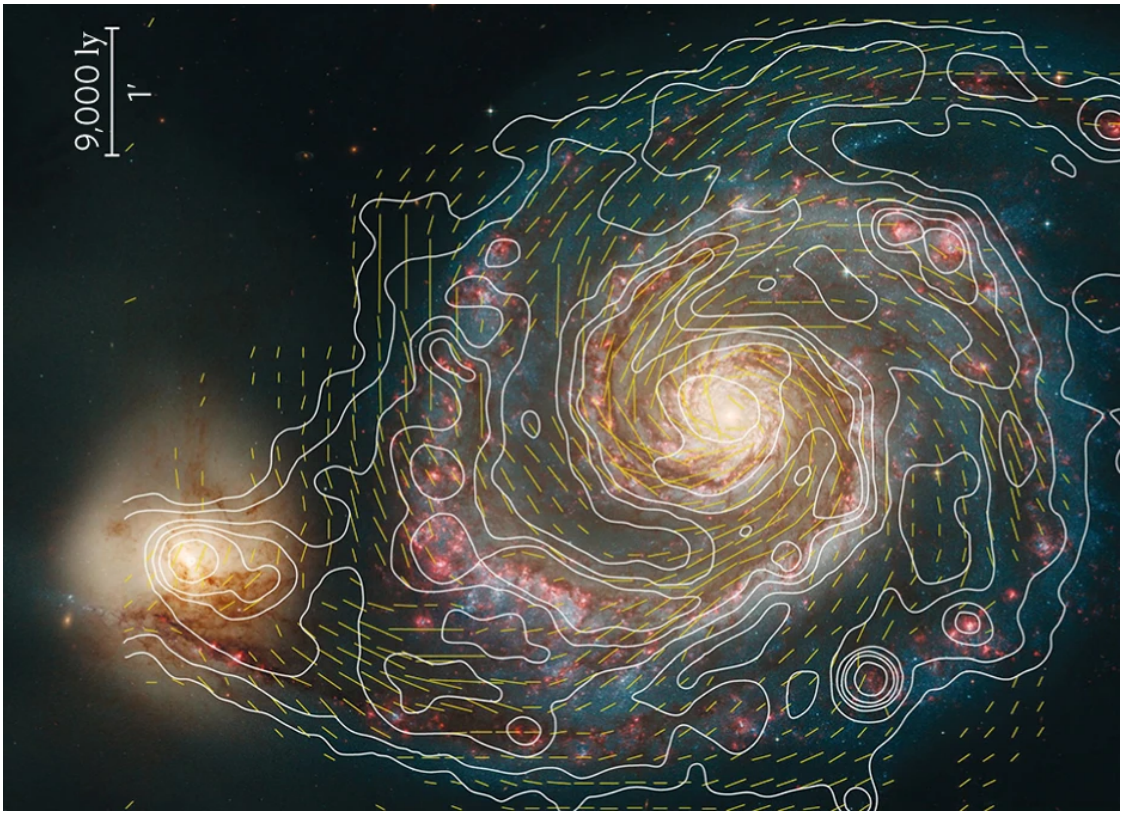


Figure 1.1: A composite optical image of the Whirlpool Galaxy (M51), combining optical data from the Hubble Space Telescope with contours of total intensity radio emission at 6 cm, with line segments tracing the magnetic field orientation.[12]

Galaxies are large, gravitationally bound systems composed of stars, stellar remnants, interstellar gas, dust, dark matter, and various other components, such as supermassive black holes, cosmic rays, molecular clouds, and active galactic nuclei

(AGN). Among their many intriguing features are the large-scale magnetic fields in the interstellar medium (ISM), with strengths typically around 10μG. While the presence of magnetic fields in the Milky Way's ISM have been known for over 60 years, the processes driving their origin and evolution remain unclear. Recent advancements in radio astronomy have enabled the estimation of magnetic field strengths and the detection of large-scale ordered fields. The Square Kilometer Array (SKA), a next-generation radio telescope, has listed "The Origin and Evolution of Cosmic Magnetism" as one of its key science projects[6]. Observational studies of galactic magnetic fields will progress even further with its advent. The success of these observational efforts will benefit greatly from theoretical improvements in understanding their origin, evolution, and properties. This work focuses on magnetic fields in disc galaxies, where observations reveal large-scale, coherent fields. A natural first question is: where do these magnetic fields come from? Are they primordial, i.e., fossil relics from the early universe, or is there an ongoing mechanism that continuously regenerates them?

To answer this, we turn to the nature of the plasmas that host these fields. These plasmas are highly turbulent. A defining feature of turbulence is the cascade of energy to smaller scales. Thus, a primordial magnetic field would struggle to survive over long timescales in high turbulence environments, as magnetic energy drains to smaller scales; ohmic dissipation is both dominant and efficient. This suggests that some mechanisms must actively sustain and even amplify these fields. That mechanism is the hydromagnetic dynamo, which refers to the process by which kinetic energy is spontaneously converted into magnetic energy.

The evolution of magnetic fields can be very well described by the equations of single-fluid magnetohydrodynamics (MHD) in most astrophysical plasmas. In this formal-

ism, the plasma behaves as a conducting fluid influenced by fluid and electromagnetic forces. However, solving the full MHD equations in turbulent regimes presents significant computational challenges due to the vast range of spatial and temporal scales involved. Thus, we rely on simplified models to study these fields. The mean-field dynamo theory is a well-accepted model that might explain the large-scale magnetic fields. This theory relies on turbulent flows breaking mirror symmetry, enabling magnetic fields to grow at scales larger than the turbulence[2][13]. Kinetic helicity is one of the most natural measures of the lack of reflectional symmetry, and it arises naturally in rotating systems[11]. The resulting large-scale magnetic fields are characterized by magnetic helicity, which measures the twist, writhe, and linkage of field lines. Kinetic and magnetic helicity are central to understanding magnetic field amplification and evolution[3]. Here, we adopt the $\alpha\Omega$ mean field dynamo model. The α-effect converts toroidal magnetic fields into poloidal fields through the action of kinetic helicity, while the Ω-effect converts poloidal fields back into toroidal fields, forming the basis of the $\alpha\Omega$-dynamo model.

However, the amplification of large-scale magnetic fields in mean-field dynamos is limited. As large-scale fields grow, small-scale magnetic helicity of opposite sign to that of the large-scale magnetic helicity is generated to conserve total magnetic helicity [3][7]. As this small-scale magnetic helicity accumulates, it reduces dynamo efficiency by suppressing the α -effect. Mechanisms to remove or redistribute small-scale helicity are crucial to maintaining dynamo activity. While resistive dissipation is too slow, processes such as large-scale outflows may expel helicity. However, in many astrophysical systems, such large outflows may not be present, which calls for alternative mechanisms to facilitate helicity transfer. Thus, the standard dynamo models may not fully capture the mechanisms driving large-scale magnetic field growth.

Despite decades of study, the connection between theory and observation in galactic magnetic field modeling remains incomplete. Standard dynamo models may not fully capture the mechanisms driving large-scale magnetic field growth. Vishniac (2012) [15] proposed a new effect arising from higher-order terms, known as the Vishniac flux—a helicity flux that could potentially dominate the dynamo process and give rise to several new properties of dynamos not predicted by conventional models.

This project aims to incorporate the Vishniac flux into the traditional mean-field dynamo model and assess its impact on the evolution of galactic magnetic fields. The analytic expression for the Vishniac flux, derived by Gopalakrishnan & Subramanian (2023), is integrated into a dynamical quenching model developed by Chamandy et al. (2014) [5]. Our model simultaneously solves the mean-field induction equation and the magnetic helicity evolution equation. In this thesis, we focus on a one-dimensional model in the vertical (z) direction, a common simplification that allows us to study the effect of a new term while retaining key dynamical features. This approach enables us to obtain the global dynamo solution at a given radial point in the galactic disc.

Unlike traditional models that primarily rely on kinetic energy, specifically the kinetic α-effect and kinetic helicity, as the main drivers of the dynamo, our model suggests that magnetic energy, in combination with large-scale rotation or shear, could be sufficient to sustain the dynamo. A central objective of this thesis is to test the hypothesis that magnetic energy, in the presence of large-scale shear or rotation, can independently sustain a galactic dynamo via the Vishniac flux. This perspective challenges foundational aspects of dynamo theory and aligns with our goal of providing a more comprehensive explanation for the observed magnetic field structures in galaxies.

Other goals of this thesis include evaluating how the inclusion of the Vishniac flux modifies the action of a conventional dynamo, for example, whether it enhances growth rates or instead suppresses dynamo activity. We also aim to determine whether there exists a parameter regime in which the Vishniac flux alone can drive a dynamo. The flux is studied both analytically, using a simplified approximation known as the no-z analysis, and numerically through simulations of the full model.

The thesis is organized as follows. In Chapter 2, we present the theoretical foundations of the mean-field dynamo formalism. Chapter 3 introduces the two turbulence profiles employed in this study. Chapters 4 and 5 detail the mathematical derivation of the Vishniac flux and a new form of the diffusive helicity flux, based on the expressions in Gopalakrishnan & Subramanian (2023), tailored to disc galaxies using the chosen turbulence profiles. Chapters 6 and 7 describe the simulation setup and specifics of the mean-field dynamo model implemented. Chapter 8 presents the simulation results and their analysis, while Chapter 9 offers analytical insights using the no-z approximation. Finally, Chapter 10 concludes the thesis with a discussion of the study's significance and possible future directions.

Chapter 2

Theoretical Background

2.1 Mean-field Induction Equation

The evolution of the magnetic field in a plasma is governed by the induction equation:

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{V} \times \mathbf{B}) + \beta \nabla^2 \mathbf{B}, \tag{2.1}$$

where $\eta = c^2/(4\pi\sigma)$ is the magnetic diffusivity. The first term in Eq. (2.1) is the induction term, representing magnetic field generation by plasma motion, and the second term is the diffusion term, representing field decay due to finite conductivity.

Using Reynolds averaging, the velocity and magnetic fields are decomposed as:

$$\mathbf{V} = \overline{\mathbf{V}} + \mathbf{v}, \quad \mathbf{B} = \overline{\mathbf{B}} + \mathbf{b}, \tag{2.2}$$

where $\overline{\mathbf{F}}$ is the mean and $\mathbf{f}$ the fluctuating component for a field $\mathbf{F}$ with $\overline{\mathbf{v}} = \overline{\mathbf{b}} = 0$.

Applying this to Eq. (2.1), the mean-field induction equation becomes:

$$\frac{\partial \overline{\mathbf{B}}}{\partial t} = \nabla \times \left(\overline{\mathbf{V}} \times \overline{\mathbf{B}} + \boldsymbol{\mathcal{E}} - \beta \nabla \times \overline{\mathbf{B}} \right), \tag{2.3}$$

where $\boldsymbol{\mathcal{E}} = \overline{\mathbf{v} \times \mathbf{b}}$ is the mean electromotive force (EMF).

The mean EMF is expressed as:

$$\boldsymbol{\mathcal{E}} = \alpha \overline{\mathbf{B}} - \eta \nabla \times \overline{\mathbf{B}}, \tag{2.4}$$

where α represents helicity-driven amplification of the magnetic field (the α-effect) and η accounts for turbulent diffusivity, which opposes field growth.

2.2 The $\alpha\Omega$-Dynamo

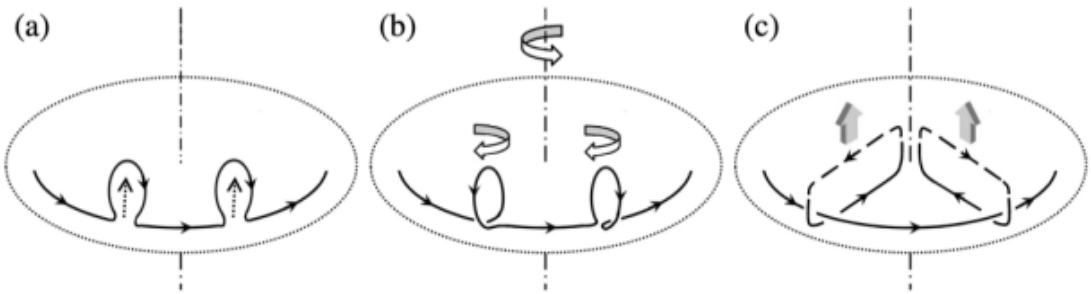


Figure 2.1: **The α-effect:** The toroidal magnetic field is first stretched into omega-shaped loops by fluid motions. These loops are then twisted by the Coriolis force, leading to the generation of a poloidal magnetic field within the disc. Only the upper half-space ($z > 0$) is shown; mirror-symmetric processes occur in the lower half-space. Adapted from [13].

The α-effect works by generating electric currents through helical turbulent motions, which interact with the toroidal magnetic field. These currents create a force that "twists" the toroidal field into a poloidal field. Essentially, the motion of the fluid moves the magnetic field lines in such a way that the circular toroidal field becomes a looped poloidal field, transferring the magnetic energy from one configuration to the other. This is how the α-effect converts a toroidal field into a poloidal field.

In disc galaxies, the mean velocity field $\overline{\mathbf{V}}$ is dominated by differential rotation. The Ω-effect arises from differential rotation (shear) in astrophysical plasmas (e.g., galactic discs, stellar interiors). It converts poloidal fields into toroidal fields, enabling a regenerative cycle when combined with the α effect.

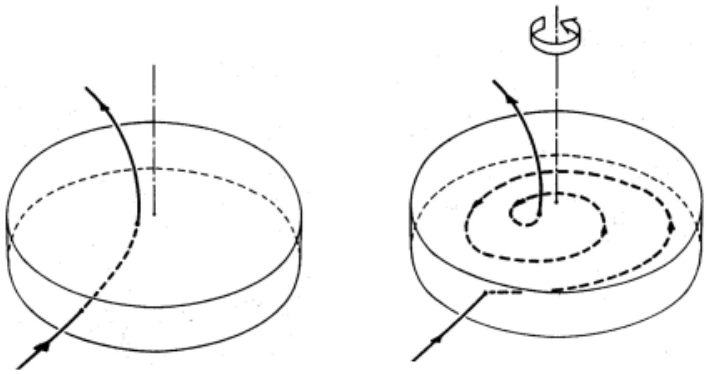


Figure 2.2: **The Ω-effect:** A poloidal magnetic field in a galactic disc is stretched by differential rotation, which is faster near the galactic center, resulting in the generation of a toroidal magnetic field. Adapted from [13].

Together, the α and Ω effects form the $\alpha\Omega$-dynamo, a positive feedback loop as shown in the figure 2.3, supporting exponential growth of the mean magnetic field ($\overline{\mathbf{B}}$) if they overcome turbulent diffusion. Without this positive feedback loop there can be no exponential growth and thus no dynamo instability .

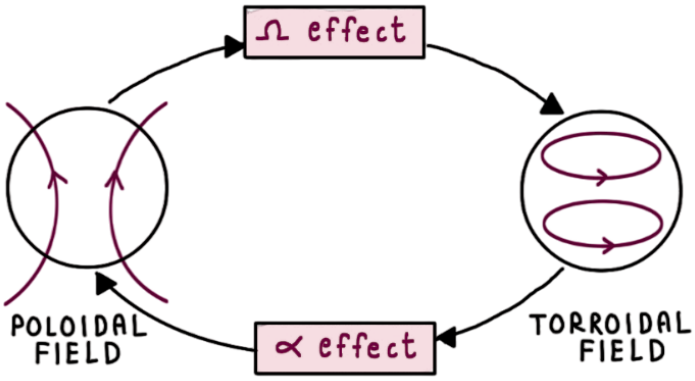


Figure 2.3: The $\alpha\Omega$ positive feedback loop.

2.3 The α-Effect

The α-effect in Equation 2.4 is represented as the sum of kinetic and magnetic contributions[5]:

$$\alpha = \alpha_k + \alpha_m, \tag{2.5}$$

where:

$$\alpha_k = -\frac{\tau}{3}\overline{\mathbf{v} \cdot (\nabla \times \mathbf{v})} \quad \text{is the kinetic part related to the mean helicity of the random flow,} \tag{2.6}$$

$$\alpha_m = \frac{\tau}{12\pi\rho}\overline{\mathbf{b} \cdot (\nabla \times \mathbf{b})} \quad \text{is the magnetic contribution.} \tag{2.7}$$

Here, ρ is the gas density, τ is the correlation time of the random flow, and $\mathbf{v}$ and $\mathbf{b}$ represent the small-scale velocity and magnetic field, respectively.

2.4 Kinematic Regime of Dynamo Action

In the kinematic regime of mean-field dynamo, the mean magnetic field is assumed to be weak enough that its feedback on the flow through the Lorentz force is negligible. This allows the velocity field to be treated as a fixed, prescribed input, ie, the system is linear. The magnetic field then evolves according to the linearized mean-field induction equation, which admits an exponential solutions of the form $\overline{\mathbf{B}}(t) \propto \exp(\gamma t)$, where γ is the growth rate of the magnetic field[13]. If $\gamma > 0$, the magnetic field grows exponentially with time, indicating dynamo action.

2.5 α-Suppression and Non-linearity

As the dynamo-generated magnetic field grows, the Lorentz force (which depends on the magnetic field) begins to significantly influence the velocity field that originally drove the dynamo. The now substantial mean magnetic field exerts a backreaction on the flow via the Lorentz force. This alters the dynamics of the turbulent velocity field, reducing the effectiveness of both the kinetic α_k-effect and the turbulent diffusivity.

When a large-scale helical magnetic field grows, it is accompanied by the generation of magnetic helicity at large scales. However, due to the approximate conservation of magnetic helicity, an equal amount of helicity with the opposite sign must build up at small scales [9][3]. The Lorentz force is dominated by small-scale contributions, leading to a small-scale current helicity that opposes the effect of helical motions by reducing the kinetic α-effect[9]. This phenomenon is known as α suppression. Without a sufficient kinetic α-effect, the positive feedback loop that drives dynamo instability breaks down, and the amplified magnetic fields decay. This process is referred to as catastrophic quenching. The system thus enters a non-linear regime where the magnetic helicity and the magnetic α_m-effect become significant, introducing a feedback mechanism.

To account for this, the mean-field dynamo model includes an additional magnetic helicity evolution equation that governs the time evolution of α_m:

$$\frac{\partial \alpha_m}{\partial t} = -\frac{2\eta}{\lambda^2 B_{\mathrm{eq}}^2} \boldsymbol{\mathcal{E}} \cdot \overline{\mathbf{B}} - \nabla \cdot \boldsymbol{\mathcal{F}}, \tag{2.8}$$

where λ is the correlation length of turbulence, and $B_{\mathrm{eq}} = \sqrt{4\pi\rho}\, v$ is the equipartition magnetic field strength corresponding to the turbulent energy density (equal to B_0 at the midplane). $\boldsymbol{\mathcal{F}}$ is the flux density of α_m, related to the flux of small-scale magnetic

helicity. The term $\nabla \cdot \boldsymbol{\mathcal{F}}$ represents helicity transport. These helicity fluxes remove the accumulated small-scale magnetic helicity from dynamo-active regions and help prevent catastrophic quenching, allowing the dynamo to reach saturation instead of decay.

2.6 Magnetic Helicity Fluxes

There are several possible contributions to the helicity flux, denoted by $\boldsymbol{\mathcal{F}}$. In general, a flux of the form

$$\boldsymbol{\mathcal{F}} = \boldsymbol{\mathcal{F}}^{NV} + \boldsymbol{\mathcal{F}}^{A} + \boldsymbol{\mathcal{F}}^{D} \tag{2.9}$$

is considered, where $\boldsymbol{\mathcal{F}}^{NV}$ is the new Vishniac flux, which is the main focus of this project, $\boldsymbol{\mathcal{F}}^{D}$ is the diffusive flux, and $\boldsymbol{\mathcal{F}}^{A}$ is the advective flux.

In this project, we do not include the advective flux; hence, only the diffusive and Vishniac fluxes are considered for helicity transport. This work presents the first numerical study of the new Vishniac flux. The detailed formulation and treatment of this flux are discussed in Chapter 4.

Chapter 3

The Turbulence Profiles

Since the form of the new Vishniac flux and the diffusive flux is sensitive to the gradients of the random velocity and magnetic field, it is important to understand how their vertical stratification (as we are working in 1D along the z-axis) affects the system. To explore this, we study two different turbulence profiles, referred to as Case 1 and Case 2.

Case 1 The profiles of the random velocity and the small-scale magnetic field are given by:

$$v^2(z) = v_0^2 \exp\left(\frac{z^2}{h^2}\right), \qquad b^2(z) = b_0^2 \exp\left(-\frac{z^2}{h^2}\right), \tag{3.1}$$

where v_0 and b_0 are the characteristic amplitudes of the turbulent velocity and magnetic field, respectively, and h is the disc scale height. In this profile, the turbulent velocity increases with distance from the midplane, resembling an inverted Gaussian distribution, and is strongest near the disc boundaries. This choice of profiles is motivated by the PhD thesis of Gressel et al. [10, see p. 123] and the textbook Astrophysical Magnetic Fields by Shukurov and Subramanian [13].

To quantify the ratio of magnetic to kinetic turbulent energies, we define the parameter:

$$\xi(z) = \frac{b^2(z)}{4\pi\rho\, v^2(z)}. \tag{3.2}$$

This can be written in the form:

$$\xi(z) = \tilde{\xi}(z)\,\xi_0, \tag{3.3}$$

where $\xi_0 = \frac{b_0^2}{4\pi\rho\, v_0^2}$ is a constant, and $\tilde{\xi}(z)$ captures the spatial dependence of the energy ratio.

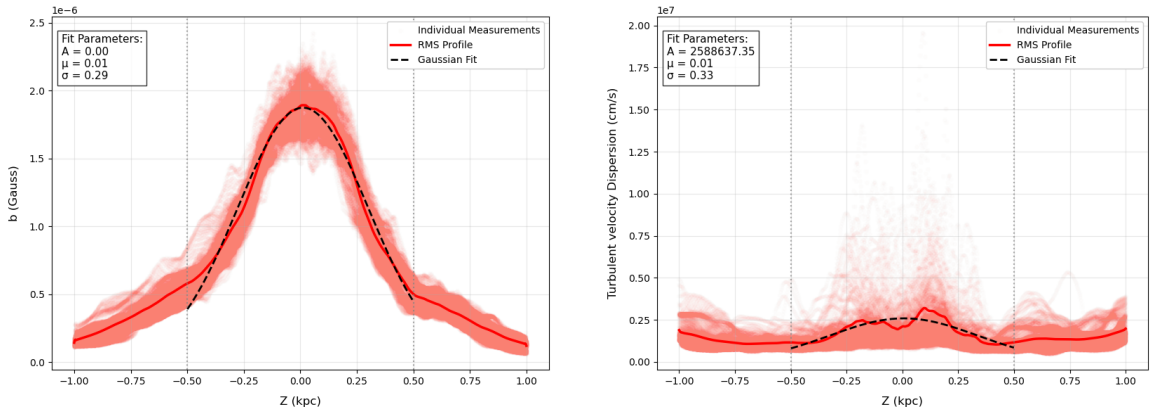


Figure 3.1: Profiles of $\boldsymbol{v}$ and $\boldsymbol{b}$ from the simulation data of Gent et al. (2024)[8], with a gaussian fitting done. We are interested in the region of Z in between the dashed lines since that is the disc height we are using in this project. We have applied gaussian fitting inside this regime. The scatter points(orange) is the full data along many realizations.

Case 2

$$v^2(z) = v_0^2 \exp\left(\frac{-z^2}{h^2}\right), \qquad b^2(z) = b_0^2 \exp\left(-\frac{z^2}{h^2}\right), \tag{3.4}$$

In case 2, the profile of random magnetic field stays the same, but we invert the velocity profile, ie, the turbulent velocity is strongest at the midplane and it weakens as we go towards the boundaries. We arrived at this choice of profiles after analysing the the simulation data of Gent et al. (2024). Through private correspondence, simulation data was obtained from Dr. Gent, including profiles of small-scale magnetic field strength and turbulent velocity across the galactic disc. These datasets were analyzed

to extract the spatial structure of the turbulence and magnetic energy. The full data span multiple time snapshots (306 timesteps) and spatial points (768 z-coordinates). We then applied an RMS averaging over time axis, because turbulence is statistically stationary in time but inhomogeneous in space - the RMS over time captures the characteristic fluctuation strength at each height (z), while spatial averaging would mix distinct physical regions. Then we did a gaussian fitting on the RMS plot to decide on the form of turbulence profile. The simulation data and gaussian fitted plots are given in figure 3.1.

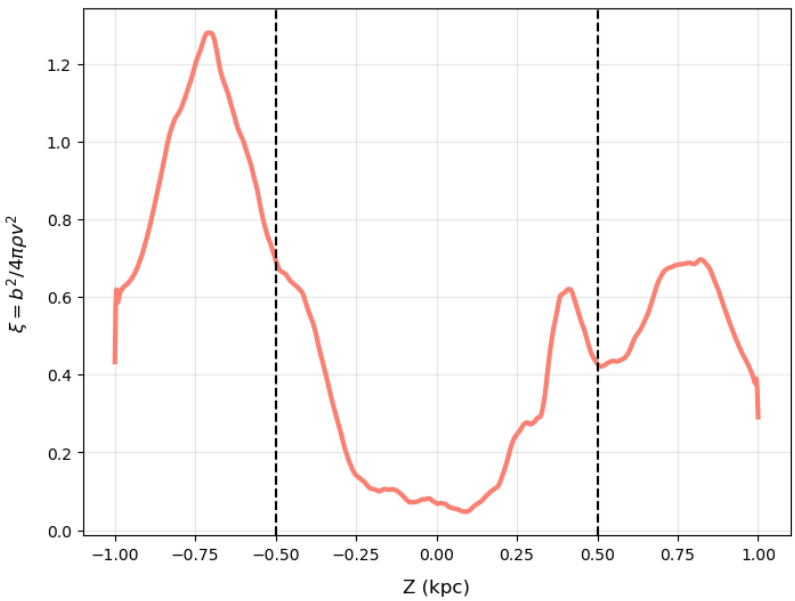


Figure 3.2: The ξ profile from simulation data[8]. We can see that it increases with z. In this profile of ξ, b, v and ρ are stratified.

The parameter ξ becomes:

$$\xi(z) = \frac{b^2(z)}{4\pi\rho\, v^2(z)} = \frac{b_0^2 \exp\left(-\frac{z^2}{h^2}\right)}{4\pi\rho\, v_0^2 \exp\left(\frac{z^2}{h^2}\right)} = \frac{b_0^2}{4\pi\rho\, v_0^2} = \xi_0. \tag{3.5}$$

Thus, in this case,

$$\xi(z) = \xi_0, \tag{3.6}$$

i.e., ξ is independent of z and is a constant. The plot of ξ obtained from simulation data [8] is shown in Figure 3.2 where we can see that ξ increases with z. This is because ρ is also a stratified quantity in that simulation[8], see Figure 3.3, but in our model, it is set as a constant as explained below, hence ξ is constant as given in Equation 3.6

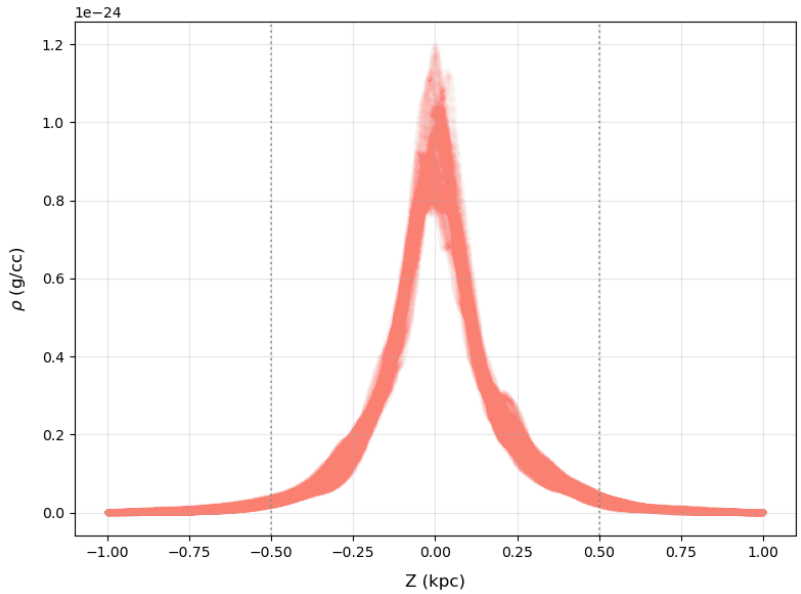


Figure 3.3: The density profile from simulation data[8]. In our simulation, density is a constant.

Density stratification We begin our derivation of the helicity fluxes in this model following the approach of Gopalakrishnan & Subramanian (2023). In their work, the fluxes are derived from higher-order correlation terms under the assumption of a constant density profile. To maintain consistency with their formalism, we also assume a uniform density ρ, independent of any vertical variation.

Chapter 4

The new Vishniac Flux

The Vishniac flux, first proposed by Dr. Ethan Vishniac in 2012[15], offers a novel perspective on dynamo theory. Unlike traditional dynamo mechanisms, this flux does not require the presence of a mean magnetic field, making it potentially influential from the earliest stages of dynamo action, when only small-scale fields and rotation are present in the system. In the context of disk galaxies, turbulence driven by supernovae in the interstellar medium, combined with small-scale magnetic fields generated by the fluctuation dynamo, provides the ideal environment for the Vishniac flux to operate and influence large-scale magnetic field evolution[9].

The Vishniac flux derived from higher order correlator terms is given by[9]:

$$\boldsymbol{\mathcal{F}}^{NV} = (\nabla\times\overline{\mathbf{V}})\cdot\left[C_1\frac{\tau}{8\pi\rho}\langle\mathbf{b}^2\rangle^2 + C_2\tau\langle\mathbf{v}^2\rangle\langle\mathbf{b}^2\rangle + C_4\lambda^2\langle\mathbf{b}^2\rangle\right]\hat{\mathbf{z}}+\tau(C_3-C_2)\left(\overline{\mathbf{V}}\times\langle\mathbf{v}^2\rangle\nabla\langle\mathbf{b}^2\rangle\right) \tag{4.1}$$

Here, $\overline{\mathbf{V}}$ is the mean velocity field, τ is the turbulent correlation time, ρ is the density, λ is the turbulent length scale, and C_1, C_2, C_3, and C_4 are constants.The values of the constants are:

$$C_1 = \frac{7}{45} \approx 0.156, \quad C_2 = -\frac{203}{5400}, \quad C_3 = \frac{403}{8100}, \quad C_4 = -\frac{1}{6}$$

Its contribution to the time evolution of the magnetic helicity density is given by the

divergence of this flux[9]:

$$-\nabla\cdot\boldsymbol{\mathcal{F}}^{NV} = -\left(\nabla\times\overline{\mathbf{V}}\right)\cdot\left[C_1\frac{\langle\mathbf{b}^2\rangle}{4\pi\rho}\tau\nabla\langle\mathbf{b}^2\rangle + C_2\langle\mathbf{b}^2\rangle\tau\nabla\langle\mathbf{v}^2\rangle + C_3\langle\mathbf{v}^2\rangle\tau\nabla\langle\mathbf{b}^2\rangle + C_4\lambda^2\nabla\langle\mathbf{b}^2\rangle\right] - \tau^2(C_3-C_2)\,\overline{\mathbf{V}}\cdot\left(\nabla\langle\mathbf{b}^2\rangle\times\nabla\langle\mathbf{v}^2\rangle\right) \tag{4.2}$$

The second term in Eq. (4.2) can be neglected if the turbulent correlators vary predominantly along a single spatial direction[9]. This condition is often satisfied in systems like disk galaxies or accretion disks[9]. The first term in Eq. (4.2), on the other hand, remains nonzero in the presence of both mean vorticity and stratification.

We consider the mean velocity field in cylindrical coordinates as:

$$\overline{\mathbf{V}} = r\Omega\,\hat{\boldsymbol{\phi}} + V_z\,\hat{\mathbf{z}} \tag{4.3}$$

To compute the curl, we use the expression for the curl in cylindrical coordinates:

$$\nabla\times\mathbf{A} = \left(\frac{1}{r}\left(\frac{\partial A_z}{\partial\phi} - \frac{\partial(rA_\phi)}{\partial z}\right)\right)\hat{\mathbf{r}} + \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right)\hat{\boldsymbol{\phi}} + \frac{1}{r}\left(\frac{\partial(rA_\phi)}{\partial r} - \frac{\partial A_r}{\partial\phi}\right)\hat{\mathbf{z}} \tag{4.4}$$

Applying this to $\overline{\mathbf{V}} = r\Omega\,\hat{\boldsymbol{\phi}} + V_z\,\hat{\mathbf{z}}$, we have:

$$A_r = 0, \quad A_\phi = r\Omega, \quad A_z = V_z$$

Radial component:

$$(\nabla\times\overline{\mathbf{V}})_r = \frac{1}{r}\left(\frac{\partial V_z}{\partial\phi} - \frac{\partial(r^2\Omega)}{\partial z}\right) \tag{4.5}$$

$$= \frac{1}{r}\frac{\partial V_z}{\partial\phi} - 2r\frac{\partial\Omega}{\partial z} \tag{4.6}$$

Azimuthal component:

$$(\nabla\times\overline{\mathbf{V}})_\phi = \frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} = -\frac{\partial V_z}{\partial r} \tag{4.7}$$

Vertical (z) component:

$$(\nabla \times \overline{\mathbf{V}})_z = \frac{1}{r}\left(\frac{\partial(r^2\Omega)}{\partial r} - \frac{\partial A_r}{\partial \phi}\right) \tag{4.8}$$

$$= \frac{1}{r}\frac{\partial(r^2\Omega)}{\partial r} \tag{4.9}$$

$$= \frac{1}{r}\left(2r\Omega + r^2\frac{\partial\Omega}{\partial r}\right) \tag{4.10}$$

$$= 2\Omega + r\frac{\partial\Omega}{\partial r} \tag{4.11}$$

Now define the shear parameter q as:

$$q = -\frac{d\ln\Omega}{d\ln r} = -\frac{r}{\Omega}\frac{d\Omega}{dr} \Rightarrow \frac{d\Omega}{dr} = -\frac{q\Omega}{r} \tag{4.12}$$

Therefore,

$$(\nabla \times \overline{\mathbf{V}})_z = 2\Omega + r\left(-\frac{q\Omega}{r}\right) = (2-q)\Omega \tag{4.13}$$

Finally, we get:

$$\nabla \times \overline{\mathbf{V}} = \left(\frac{1}{r}\frac{\partial V_z}{\partial \phi} - 2r\frac{\partial\Omega}{\partial z}\right)\hat{\mathbf{r}} - \frac{\partial V_z}{\partial r}\hat{\boldsymbol{\phi}} + (2-q)\Omega\hat{\mathbf{z}} \tag{4.14}$$

Under the assumptions of axisymmetry and the slab approximation, as detailed in Chapter 7, only the $\hat{\mathbf{z}}$ component remains in the curl of the velocity field. Note that we are not considering any vertical shear. Thus, we have:

$$\nabla \times \overline{\mathbf{V}} = (2-q)\Omega\,\hat{\mathbf{z}} \tag{4.15}$$

Now, as discussed in Chapter 3, we consider two different cases of turbulence profiles and derive the final form of the Vishniac flux that contributes to the evolution equation for α_m.

4.1 Case 1

Here, we assume the turbulence profiles to be:

$$b(z) = b_0 \exp\left(-\frac{z^2}{2h^2}\right), \tag{4.16}$$

$$v(z) = v_0 \exp\left(\frac{z^2}{2h^2}\right), \tag{4.17}$$

where ρ is kept constant. The parameter ξ becomes:

$$\xi(z) = \frac{b^2}{4\pi\rho v^2} = \tilde{\xi}(z)\,\xi_0, \tag{4.18}$$

where $\xi_0 = \frac{b_0^2}{4\pi\rho v_0^2}$ is a constant.

Now, consider the first term in the brackets of Equation (4.2):

$$\text{1st term} = C_1 \frac{\langle b^2 \rangle}{4\pi\rho} \tau \nabla \langle b^2 \rangle \tag{4.19}$$

$$= C_1 \xi_0^2 (4\pi\rho) v_0^4 \tau \left(-\frac{2z}{h^2}\right) \exp\left(-\frac{2z^2}{h^2}\right) \hat{\mathbf{z}}. \tag{4.20}$$

Simplifying the remaining terms:

$$\text{Remaining terms} = (C_2 - C_3 - C_4) b_0^2 \tau v_0^2 \left(\frac{2z}{h^2}\right) \hat{\mathbf{z}} \tag{4.21}$$

$$= 4\pi\rho v_0^4 \xi_0 \tau \left(\frac{2z}{h^2}\right) (C_2 - C_3 - C_4) \hat{\mathbf{z}}. \tag{4.22}$$

Here, we have used $\lambda = v\tau$. Now, combining all terms:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = -(2-q)\Omega\, 4\pi\rho\tau v_0^4 \xi_0 \left(\frac{2z}{h^2}\right) \left[C_2 - C_3 - C_4 - C_1 \xi_0 \exp\left(-\frac{2z^2}{h^2}\right)\right]. \tag{4.23}$$

We now introduce the dimensionless parameter R_Ω, which controls Ω through the relation:

$$R_\Omega = \frac{-q\Omega h^2}{\eta}, \tag{4.24}$$

where η is the turbulent diffusivity, given by:

$$\eta = \frac{1}{3}\tau v^2. \tag{4.25}$$

Rearranging for Ω, we get:

$$\Omega = \frac{-R_\Omega \eta}{qh^2}. \tag{4.26}$$

Substituting this into Equation (4.23), we obtain:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = (2-q)\frac{8\pi\rho R_\Omega \eta \tau v_0^4 \xi_0}{q}\left(\frac{z}{h^4}\right)\left[0.0793 - 0.1556\xi_0 \exp\left(-\frac{2z^2}{h^2}\right)\right]. \tag{4.27}$$

This is the final expression for the Vishniac flux in Case 1 of the turbulence profiles considered in our simulation.

4.2 Case 2

The turbulence profiles in this case are:

$$b(z) = b_0 \exp\left(-\frac{z^2}{2h^2}\right), \tag{4.28}$$

$$v(z) = v_0 \exp\left(-\frac{z^2}{2h^2}\right). \tag{4.29}$$

Here, ξ becomes:

$$\xi_0 = \frac{b_0^2}{4\pi\rho v_0^2} = \xi \tag{4.30}$$

As in Case 1, the density ρ is assumed to be constant. The only change from Case 1 is in the velocity profile. Consequently, in Equation (4.2), the C_1 term remains the same as before. However, the remaining terms that involve the gradient of the velocity profile acquire a negative sign and an additional factor of $\exp\left(-\frac{2z^2}{h^2}\right)$.

Following the same steps as detailed in Case 1, we obtain:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = (2-q)\frac{4\pi\rho R_{\Omega}\eta\tau v^4\xi}{q}\left(\frac{2z}{h^4}\right)[0.1545 - 0.1556\xi]\exp\left(-\frac{2z^2}{h^2}\right). \quad (4.31)$$

This is the final expression for the Vishniac flux corresponding to Case 2 of the turbulence profiles considered in our simulation.

Chapter 5

The new Diffusive Flux

Diffusive fluxes of magnetic helicity have been incorporated into models of the mean-field dynamo that include a dynamical equation for the current helicity of the small-scale magnetic fields. Such fluxes have been measured in direct numerical simulations. Earlier works employed a heuristic Fickian diffusion expression, given by:

$$\boldsymbol{\mathcal{F}}^D = -\kappa \nabla \alpha_m, \tag{5.1}$$

where $\kappa \approx 0.3\eta$, as obtained from numerical simulations.

In Gopalakrishnan & Subramanian (2023), the flux density was derived more rigorously by considering higher-order terms in the magnetohydrodynamic equations. The newer formulation of the diffusive helicity flux is grounded in the physical processes underlying turbulence and the evolution of magnetic helicity.

The derived form is as follows[9]:

$$\boldsymbol{\mathcal{F}}^D = -\left(\frac{7\tau}{27}\langle \boldsymbol{v}^2\rangle + \frac{7\tau}{27}\frac{\langle \boldsymbol{b}^2\rangle}{4\pi\rho}\right)\nabla h^b \tag{5.2}$$

Taking the divergence:

$$-\nabla\cdot\boldsymbol{\mathcal{F}}^D = \nabla\cdot\left(\frac{7\tau}{27}\langle \boldsymbol{v}^2\rangle \nabla h^b\right) + \nabla\cdot\left(\frac{7\tau}{27}\frac{\langle \boldsymbol{b}^2\rangle}{4\pi\rho}\nabla h^b\right) \tag{5.3}$$

$$= \frac{7\tau}{27}\left[\langle \boldsymbol{v}^2\rangle \nabla^2 h^b + \nabla\langle \boldsymbol{v}^2\rangle\cdot\nabla h^b\right] + \frac{7\tau}{27}\left[\frac{\langle \boldsymbol{b}^2\rangle}{4\pi\rho}\nabla^2 h^b + \nabla\left(\frac{\langle \boldsymbol{b}^2\rangle}{4\pi\rho}\right)\cdot\nabla h^b\right] \tag{5.4}$$

Group into two terms:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^D = \underbrace{\left(\frac{7\tau}{27}\langle \boldsymbol{v}^2\rangle + \frac{7\tau}{27}\frac{\langle \boldsymbol{b}^2\rangle}{4\pi\rho}\right)\nabla^2 h^b}_{\text{Term 1}} + \underbrace{\left(\frac{7\tau}{27}\nabla\langle \boldsymbol{v}^2\rangle + \frac{7\tau}{27}\nabla\left(\frac{\langle \boldsymbol{b}^2\rangle}{4\pi\rho}\right)\right)\cdot\nabla h^b}_{\text{Term 2}} \tag{5.5}$$

5.1 Case 1

Term 1 Plug in the profiles:

$$\text{Term 1} = \frac{7\tau}{27}\left[v_0^2 \exp\left(\frac{z^2}{h^2}\right) + \frac{b_0^2}{4\pi\rho}\exp\left(-\frac{z^2}{h^2}\right)\right]\nabla^2 h^b \tag{5.6}$$

$$\text{Term 1} = \frac{7\tau v_0^2}{27}\left[\exp\left(\frac{z^2}{h^2}\right) + \xi_0 \exp\left(-\frac{z^2}{h^2}\right)\right]\nabla^2 h^b \tag{5.7}$$

Term 2

$$\text{Term 2} = \frac{7\tau}{27}\left[\nabla\langle \boldsymbol{v}^2\rangle + \frac{1}{4\pi\rho}\nabla\langle \boldsymbol{b}^2\rangle\right]\cdot\nabla h^b \tag{5.8}$$

$$\text{Term 2} = \frac{7\tau}{27}\left[\frac{2z}{h^2}v_0^2 \exp\left(\frac{z^2}{h^2}\right) - \frac{2z}{h^2}\frac{b_0^2}{4\pi\rho}\exp\left(-\frac{z^2}{h^2}\right)\right]\hat{z}\cdot\nabla h^b \tag{5.9}$$

$$= \left[v_0^2 \exp\left(\frac{z^2}{h^2}\right) - \frac{b_0^2}{4\pi\rho}\exp\left(-\frac{z^2}{h^2}\right)\right]\cdot\frac{14z\tau}{27h^2}\,\partial_z h^b \tag{5.10}$$

$$\text{Term 2} = \left[\exp\left(\frac{z^2}{h^2}\right) - \xi_0 \exp\left(-\frac{z^2}{h^2}\right)\right]\cdot\frac{14zv_0^2\tau}{27h^2}\,\partial_z h^b \tag{5.11}$$

Final Expression for $-\nabla \cdot \boldsymbol{\mathcal{F}}^D$

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^D = \frac{7\tau v_0^2}{27}\left[\exp\left(\frac{z^2}{h^2}\right) + \xi_0 \exp\left(-\frac{z^2}{h^2}\right)\right]\nabla^2 h^b \tag{5.12}$$

$$+ \frac{14 z v_0^2 \tau}{27 h^2}\left[\exp\left(\frac{z^2}{h^2}\right) - \xi_0 \exp\left(-\frac{z^2}{h^2}\right)\right]\partial_z h^b \tag{5.13}$$

Let $h^b = \langle a \cdot b \rangle$

From Sur et al. (2007)[14], the evolution equation for $\chi = h^b$ is:

$$\frac{\partial \boldsymbol{\chi}}{\partial t} + \nabla \cdot \boldsymbol{\mathcal{F}} = -2\boldsymbol{\mathcal{E}} \cdot \boldsymbol{\mathcal{B}} - 2\eta \boldsymbol{\mathcal{J}} \cdot \boldsymbol{\mathcal{B}} \tag{5.14}$$

$$\alpha_m = \frac{1}{3}\tau \frac{1}{l_0^2}\frac{\chi}{\rho}\left(\frac{1}{4\pi}\right) \tag{5.15}$$

Assuming $\chi = h^b$ and $l_0 = v\tau$, we get:

$$h^b = \frac{3\rho l_0^2}{\tau}\alpha_m(4\pi) = \frac{3\rho\tau^2 v^2}{\tau}\alpha_m(4\pi) = 3\rho\tau v^2 \alpha_m(4\pi) \tag{5.16}$$

Gradient of h^b We now compute:

$$\nabla h^b = \nabla\left(3\rho\tau v^2 \alpha_m \cdot 4\pi\right) \tag{5.17}$$

$$= 3\rho\tau \cdot 4\pi\left(\nabla(v^2\alpha_m)\right) \tag{5.18}$$

$$= 3\rho\tau \cdot 4\pi\left(v^2\nabla\alpha_m + \alpha_m \nabla v^2\right) \tag{5.19}$$

Assuming vertical stratification along z, and

$$v^2 = v_0^2 \exp\left(\frac{z^2}{h^2}\right), \quad \Rightarrow \nabla v^2 = \frac{2z}{h^2} v^2 \hat{z} \tag{5.20}$$

Then:

$$\nabla h^b = 3\rho\tau \cdot 4\pi \left[v_0^2 \exp\left(\frac{z^2}{h^2}\right) \nabla \alpha_m + \alpha_m \cdot \frac{2z}{h^2} v_0^2 \exp\left(\frac{z^2}{h^2}\right) \hat{z} \right] \tag{5.21}$$

$$= 3\rho\tau v_0^2 \cdot 4\pi \exp\left(\frac{z^2}{h^2}\right) \left[\nabla \alpha_m + \alpha_m \cdot \frac{2z}{h^2} \hat{z} \right] \tag{5.22}$$

Laplacian of h^b Now:

$$\nabla^2 h^b = 3\rho\tau \cdot 4\pi \nabla^2 (v^2 \alpha_m) \tag{5.23}$$

$$= 3\rho\tau \cdot 4\pi \left[2\nabla v^2 \cdot \nabla \alpha_m + \alpha_m \nabla^2 v^2 + v^2 \nabla^2 \alpha_m \right] \tag{5.24}$$

We compute:

$$\nabla v^2 = \frac{2z}{h^2} v^2 \hat{z}, \quad \nabla^2 v^2 = \frac{2}{h^2} v^2 + \frac{4z^2}{h^4} v^2 \tag{5.25}$$

So:

$$\nabla^2 h^b = 3\rho\tau \cdot 4\pi \left[2 \cdot \frac{2z}{h^2} v^2 \partial_z \alpha_m + \alpha_m \left(\frac{2}{h^2} v^2 + \frac{4z^2}{h^4} v^2 \right) + v^2 \nabla^2 \alpha_m \right] \tag{5.26}$$

$$= 3\rho\tau v^2 \cdot 4\pi \left[\frac{4z}{h^2} \partial_z \alpha_m + \left(\frac{2}{h^2} + \frac{4z^2}{h^4} \right) \alpha_m + \nabla^2 \alpha_m \right] \tag{5.27}$$

Thus, the final expression for the modified form of the diffusive flux employed in this project is given in Equation 5.12. We substitute the forms of Equations 5.26 and 5.21 into this expression to obtain the complete flux formulation.

5.2 Case 2

The turbulence profiles for Case 2 are identical to those in Case 1, except for the velocity profile. The steps of the derivation remain exactly the same as in Case 1. In this case, Equation 5.2 acquires a total multiplicative factor of:

$$\left\langle v^2 \right\rangle + \frac{\langle b^2 \rangle}{4\pi\rho} = v_0^2(1+\xi_0)\exp\left(-\frac{z^2}{h^2}\right). \tag{5.28}$$

Proceeding term by term, as done previously:

Term 1

$$\text{Term 1} = \frac{7\tau}{27}\left(\left\langle v^2 \right\rangle + \frac{\langle b^2 \rangle}{4\pi\rho}\right)\nabla^2 h^b \tag{5.29}$$

$$= \frac{7\tau v_0^2}{27}(1+\xi_0)\exp\left(-\frac{z^2}{h^2}\right)\nabla^2 h^b. \tag{5.30}$$

Term 2

$$\text{Term 2} = \frac{7\tau}{27}\left[\nabla\left\langle v^2 \right\rangle + \frac{1}{4\pi\rho}\nabla\left\langle b^2 \right\rangle\right]\cdot\nabla h^b \tag{5.31}$$

$$= -\frac{14z\tau}{27h^2}v_0^2(1+\xi_0)\exp\left(-\frac{z^2}{h^2}\right)\partial_z h^b. \tag{5.32}$$

Final Expression

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^D = \frac{7\tau v_0^2}{27}(1+\xi_0)\exp\left(-\frac{z^2}{h^2}\right)\left[\nabla^2 h^b - \frac{2z}{h^2}\partial_z h^b\right]. \tag{5.33}$$

Gradient of h^b The velocity profile is given by:

$$v^2 = v_0^2 \exp\left(-\frac{z^2}{h^2}\right) \quad \Rightarrow \quad \nabla v^2 = -\frac{2z}{h^2} v^2\,\hat{z}. \tag{5.34}$$

Thus, the gradient of h^b becomes:

$$\nabla h^b = 3\rho\tau v_0^2 \cdot 4\pi \exp\left(-\frac{z^2}{h^2}\right)\left[\nabla\alpha_m - \alpha_m \cdot \frac{2z}{h^2}\hat{z}\right]. \tag{5.35}$$

Laplacian of h^b The Laplacian of h^b is computed as follows:

$$\nabla^2 h^b = 3\rho\tau \cdot 4\pi \nabla^2(v^2\alpha_m) = 3\rho\tau \cdot 4\pi\left[2\nabla v^2 \cdot \nabla\alpha_m + \alpha_m \nabla^2 v^2 + v^2 \nabla^2 \alpha_m\right]. \tag{5.36}$$

With:

$$\nabla v^2 = -\frac{2z}{h^2} v^2\,\hat{z}, \quad \nabla^2 v^2 = \left(-\frac{2}{h^2} + \frac{4z^2}{h^4}\right) v^2, \tag{5.37}$$

we obtain:

$$\nabla^2 h^b = 3\rho\tau v_0^2 \cdot 4\pi \exp\left(-\frac{z^2}{h^2}\right)\left[\nabla^2\alpha_m - \frac{4z}{h^2}\partial_z\alpha_m + \left(-\frac{2}{h^2} + \frac{4z^2}{h^4}\right)\alpha_m\right]. \tag{5.38}$$

For the diffusive flux derived using the Case 2 turbulence profiles, we use the form given in Equation 5.33, and substitute into it the expressions for the gradient and Laplacian of h^b from Equations 5.35 and 5.38, respectively.

Chapter 6

The Simulation Setup

The dynamical quenching model, ie, mean field induction equation and helicity evolution equation, are solved numerically using a simulation code developed in FORTRAN (Fortran 95). The numerical schemes are inspired by the well-established Pencil Code, which employs a high-order finite-difference, non-conservative formulation for compressible magnetohydrodynamic flows; the time integration in it is implemented by an explicit scheme.

Inspired from this I developed the simulation using FORTRAN 95. Spatial derivatives are approximated using a sixth-order finite differencing method, which balances accuracy and efficiency. Higher-order schemes (8th and 12th) showed only marginal gains in accuracy but were more computationally demanding, requiring increased memory and longer run times. Hence, the sixth-order method was chosen as the most efficient solution.

Time evolution is handled using a third-order implicit Runge–Kutta (RK3) scheme [4]. Although a fourth-order explicit RK4 scheme was also tested, no significant difference in the results was observed. A key advantage of the implicit RK3 over the explicit RK4 scheme is its stability when handling the stiff equations that arise in MHD simulations, where processes like diffusion and advection operate on different timescales. The implicit RK3 scheme enables larger time steps while maintaining stability, making it more computationally efficient than the RK4 scheme, which requires

significantly smaller time steps, thereby increasing memory usage and run time to maintain stability.

The simulation employs a spatial resolution of 101 grid points. While models were also tested with 51 and 201 grid points, the 201-grid resolution, though providing slightly more accurate results, was significantly more time-consuming. Therefore, a resolution of 101 grid points was used, and every new model was also tested with 201 grid points to see if there are any pronounced differences or new structures in the output.

Boundary conditions are implemented using ghost zones, which are additional grid points beyond the physical domain of the simulation. These ghost zones are not part of the actual solution, but are used to impose boundary conditions by mirroring, extrapolating, or applying specific rules to the data inside the physical grid. In the code, three ghost cells are placed on each side of the physical grid, resulting in a total of 107 grid points. The use of ghost zones is important for maintaining the accuracy of the finite differencing scheme near the boundaries, as they allow for consistent calculation of derivatives and prevent numerical instabilities that might arise from directly enforcing boundary conditions.

Chapter 7

The Model

7.1 Assumptions

We adopt a cylindrical coordinate system (r, ϕ, z) and solve the induction equation (Equation 2.3) under the assumption of axisymmetry, which neglects azimuthal (ϕ) derivatives, ie, our system is independent of ϕ.

Applying the thin-disc approximation, the vertical magnetic field component is considered negligible compared to the radial and azimuthal components, i.e., $|\overline{B}_z| \ll |\overline{B}_r|, |\overline{B}_\phi|$, and radial (r) derivatives are neglected. These simplifications reduce the problem to a one-dimensional form in z, allowing local dynamo solutions. We solve for $\overline{B}_r$, $\overline{B}_\phi$, and α_m at $r = 4$ kpc. Using $\nabla \cdot \overline{\mathbf{B}} = 0$, the vertical magnetic field $\overline{B_z}$ is determined.

The reduced induction equations for the $\alpha\Omega$ dynamo are:

$$\frac{\partial \overline{B}_r}{\partial t} = -\frac{\partial}{\partial z}\left(\alpha \overline{B}_\phi\right) + \eta \frac{\partial^2 \overline{B}_r}{\partial z^2} \tag{7.1}$$

$$\frac{\partial \overline{B}_\phi}{\partial t} = -\Omega \overline{B}_r + \eta \frac{\partial^2 \overline{B}_\phi}{\partial z^2} \tag{7.2}$$

The α_m evolution equation given in equation 2.8 becomes:

$$\frac{\partial \alpha_m}{\partial t} = -\frac{2\eta}{l^2 B_{\mathrm{eq}}^2}\left[\alpha\left(\overline{B}_r^2 + \overline{B}_\phi^2\right) - \eta\left(\frac{\partial \overline{B}_r}{\partial z}\overline{B}_\phi - \frac{\partial \overline{B}_\phi}{\partial z}\overline{B}_r\right)\right] - \nabla \cdot \boldsymbol{\mathcal{F}}^{NV} - \nabla \cdot \boldsymbol{\mathcal{F}}^{D}, \tag{7.3}$$

The magnetic pitch angle is the angle between the azimuthal and radial components

of the magnetic field in a disk, indicating the spiral structure of the field:

$$p = \arctan\left(\frac{\overline{B}_r}{\overline{B}_\phi}\right), \quad p \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \tag{7.4}$$

In a thin disc, the α-effect is written as:

$$\alpha_k = \alpha_0(r)\tilde{\alpha}(z), \tag{7.5}$$

where $\alpha_0 = \frac{l^2\Omega}{h}$ with h as the disc half-thickness. From symmetry considerations, α is an odd function of z, a conventional choice of the function is[5]:

$$\tilde{\alpha}(z) = \sin\left(\frac{\pi z}{h}\right). \tag{7.6}$$

We consider specific models at $r = 4\,\mathrm{kpc}$. Dimensionless control parameters are defined as:

$$R_\alpha \equiv \frac{\alpha_0 h}{\eta}, \quad R_\Omega \equiv \frac{-q\Omega h^2}{\eta} \tag{7.7}$$

Lengths are scaled to the disk half-thickness h, except for radial measurements, which are scaled to the disk radius R. The magnetic field is scaled to $B_0 \sim 1\,\mu\mathrm{G}$.

7.2 Initial Conditions

The induction equation is homogeneous, and $\mathbf{B} = 0$ is a valid solution that satisfies both homogeneous initial and boundary conditions. Consequently, in the absence of an initial magnetic field, no magnetic field would be generated through standard (single-fluid) MHD processes. Therefore, hydromagnetic dynamos require a non-zero seed magnetic field to initiate field amplification.

In mean-field dynamo simulations, the specific form of the seed field is generally not critical, as the system typically loses memory of its initial state due to the exponential

growth (or decay) of magnetic modes. In this study, the simulation is initialized with a weak seed magnetic field having a Gaussian profile and an amplitude of 10^{-2}.

7.3 Boundary Conditions

For boundary conditions, the standard vacuum boundary conditions are employed, where the region outside the dynamo-active disk is assumed to be a perfect electrical insulator. In such a vacuum, there are no mean electric currents, so the curl of the magnetic field vanishes:

$$\nabla \times \mathbf{B} = 0 \quad \text{for } |z| > h. \tag{7.8}$$

This implies that the magnetic field outside the disc is potential, and can therefore be expressed as the gradient of a scalar potential Φ:

$$\mathbf{B} = \nabla \Phi. \tag{7.9}$$

In an axisymmetric system, Φ must be independent of the azimuthal coordinate. As a result, the magnetic field component $\overline{B_\phi} = 0$. Additionally, since $\mathbf{B} = \nabla\Phi$, the radial component is given by:

$$B_r = \frac{\partial \Phi}{\partial r}, \quad \text{for } |z| > h. \tag{7.10}$$

Now, when we neglect the radial derivatives using thin-disc approximation, we get:

$$B_r = 0 \quad \text{for } |z| > h. \tag{7.11}$$

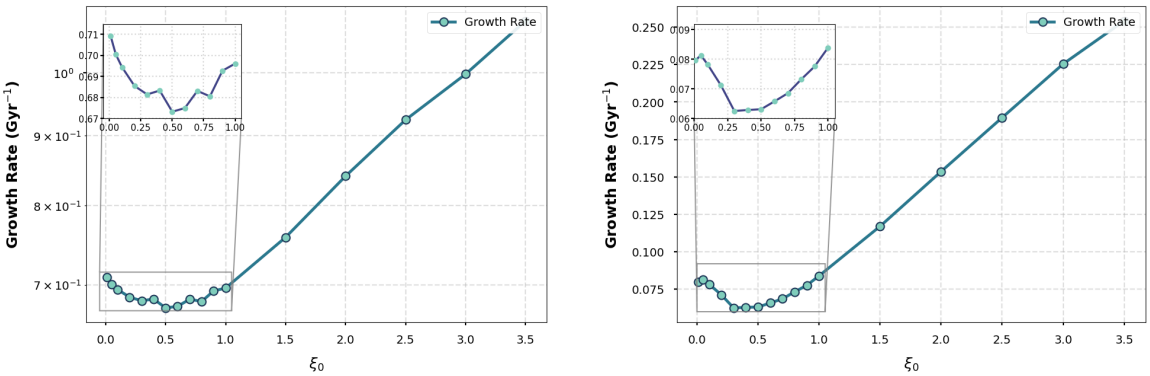
In the resulting magnetic field above and below the disc is purely vertical, our boundary condition is thus:

$$\overline{B}_r = \overline{B}_\phi = 0, \quad \alpha_m = 0, \quad \text{at } z = \pm h. \tag{7.12}$$

Chapter 8

Results and Discussion

8.1 The Vishniac Flux in Mean-Field Dynamos: Aid or Hindrance?



(a) Case 1

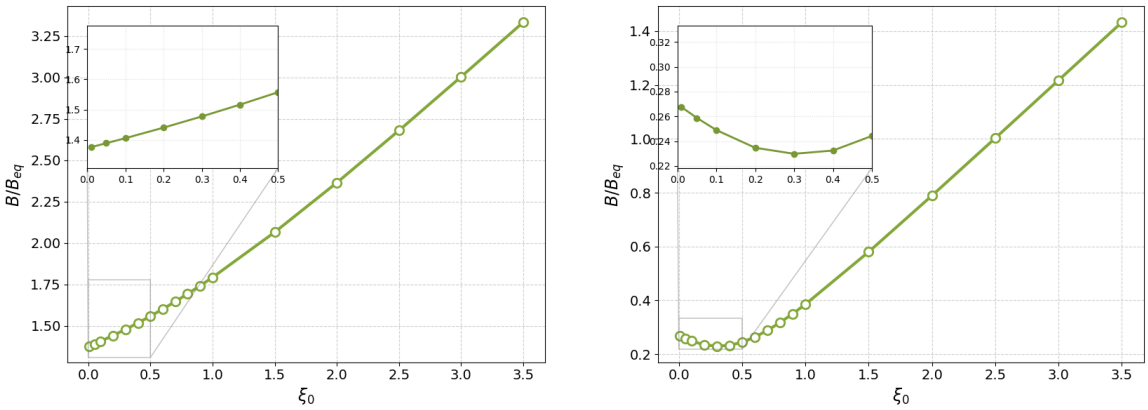
(b) Case 2

Figure 8.1: Growth rate values for Case 1 (left) and Case 2 (right) across different ξ_0 values. All tested models include the kinetic α-effect (α_k). Case 1 and Case 2 correspond to the different velocity profiles discussed in Chapter 3.

We carried out a parameter sweep over ξ_0, keeping $v_0 = 20$ and $R_\Omega = -20$. This was done to assess whether the Vishniac flux helps or hinders dynamo action, and to check if there exists a critical range of ξ_0 values where it begins to contribute positively. The plots of growth rate vs. ξ_0 and saturation strength vs. ξ_0 for both Case 1 and Case 2 are shown in the figures.

We varied ξ_0 from 0.01 to 3.5. Although such large values of ξ_0 are not realistic in astrophysical contexts, they were intentionally included to explore the theoretical

behavior of the model. In realistic astrophysical settings, ξ_0 is expected to be much lower; however, it is at these higher values that the Vishniac flux begins to significantly enhance dynamo action.



(a) Case 1 (b) Case 2

Figure 8.2: Magnetic field saturation strength for Case 1 (left) and Case 2 (right) across different ξ_0 values. All tested models include the kinetic α-effect (α_k).

In Figures 8.1 and 8.2, we show how the growth rate and the final saturation strength of the magnetic field vary as ξ_0 ranges from 0.01 to 3.5. For Case 1, it is evident that below a certain critical ξ_0, the Vishniac flux suppresses the dynamo by reducing the growth rate. This behavior aligns with the derived expression for the Vishniac flux in Case 1.

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} \propto \left[0.079 - 0.156\xi_0 e^{-2z^2/h^2}\right] \tag{8.1}$$

From this, we observe that there is a critical value of ξ_0 at which the sign of the flux changes. This value can be calculated numerically and is found to be $\xi_0 = 0.506$. In Figures 8.1 and 8.2, the effect of this transition point is clearly visible. For $\xi_0 < 0.506$, the flux is negative and contributes detrimentally to the evolution of α_m. It essentially counteracts the driving term provided by α_k. In simulations without α_k and for

$\xi_0 < 0.506$, the magnetic field decays, and the decay rate slows as ξ_0 approaches 0.506. This trend aligns with expectations based on the midplane analysis.

Once $\xi_0 > 0.506$, the flux flips sign and begins to aid the dynamo. This is evident from the sharp increase in growth rate beyond $\xi_0 = 0.5$ in Figure 8.1. For $\xi_0 > 1$, where the small-scale fields become super-equipartition and begin to dominate, the flux enhances the growth rate beyond that of the conventional dynamo. This suggests that at sufficiently large ξ_0, the Vishniac term could potentially dominate the helicity evolution and drive the dynamo on its own. This point is discussed further in the next section.

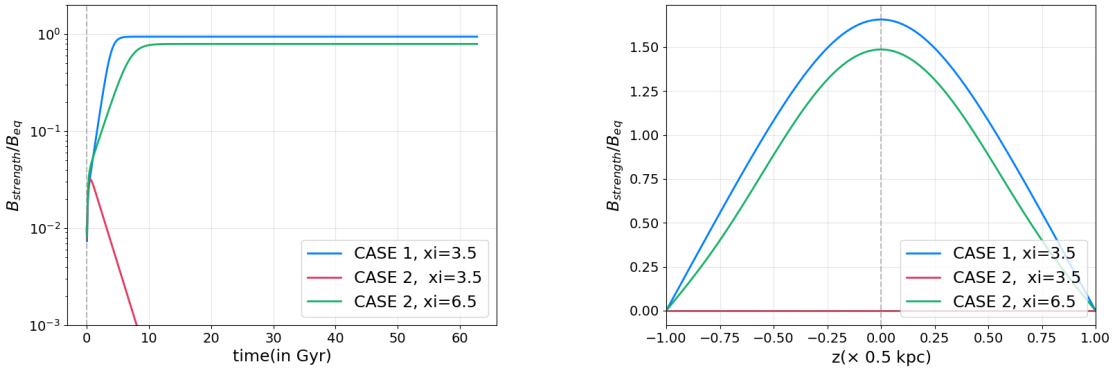
Now, considering Case 2, the flux takes the form:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} \propto [0.154 - 0.156\xi_0]\, e^{-2z^2/h^2} \tag{8.2}$$

Here, the critical value where the flux flips sign is $\xi_0 = 0.987$. Additionally, there is an exponential factor outside the bracket in Case 2, which causes the flux to resemble a bell curve peaking at the midplane. This means that the Vishniac term has a highly localized effect in this case, with most of the activity concentrated near the midplane.

From the simulations (see Figure 8.1), we observe that the overall trend in Case 2 is similar to that of Case 1. However, the increase in growth rate with increasing ξ_0 is slower. Even at $\xi_0 = 3.5$, the growth rate does not exceed that of the conventional dynamo. This suggests that unless ξ_0 becomes very large, the flux remains too localized to be significantly beneficial. For the flux to have a notable effect, ξ_0 must be high enough for its influence to extend beyond just the midplane.

8.2 The Vishniac Flux Action Without α_k Driving the Dynamo



(a) Mean field strength vs. time. The y-axis is shown in logarithmic scale.

(b) Total field strength vs. z. This is the vector sum of $\overline{B}_\phi$ and $\overline{B}_r$.

Figure 8.3: Dynamo models with a purely magnetic α-effect ($\alpha = \alpha_m$), excluding the conventional kinetic contribution (α_k). Case 1 and Case 2 correspond to the different velocity profiles discussed in Chapter 3.

As shown in the previous section, there is evidence that at super-equipartition values of ξ_0, the Vishniac flux alone can sustain dynamo action. To investigate this possibility, we considered three models. Two of these are Case 1 and Case 2 evaluated with $\xi_0 = 3.5$. In both cases, we explicitly turned off the kinetic helicity term α_k, meaning the usual driver of large-scale field growth was absent.

Surprisingly, in Case 1, we still observed a growing and eventually saturated magnetic field—driven purely by the Vishniac flux. In Case 2, the field initially decayed for the same ξ_0 value. However, based on the trend seen in Figure 8.1, we can infer that even Case 2 could support dynamo action if ξ_0 were increased further. This was confirmed: at higher ξ_0, Case 2 also exhibited self-sustained field growth.

But how is this possible, given that $\alpha_k = 0$? In the conventional picture of a galactic

dynamo, turbulence (e.g., from supernova explosions) produces kinetic helicity, which breaks reflectional symmetry and gives rise to α_k. This, in combination with differential rotation (Ω), forms the classic $\alpha\Omega$ dynamo loop: poloidal fields generate toroidal fields and vice versa, leading to exponential field growth. When α_k is switched off, that feedback loop is broken, and we expect no sustained dynamo.

However, the Vishniac flux introduces a twist to this picture. In the standard scenario, the role of the flux is to avoid catastrophic α-quenching by transporting small-scale magnetic helicity out of the dynamo-active region. But at sufficiently large ξ_0, this flux not only removes excess helicity—it also acts as a driving term in the magnetic helicity evolution equation.

This can be explained using the evolution equation for magnetic α. Assume a high-ξ_0 case where the flux has an overall positive sign and is larger than the diffusive term:

$$\frac{\partial \alpha_m}{\partial t} \approx -\frac{2\eta}{l^2 B_{\mathrm{eq}}^2}\left[\alpha\left(\overline{B}_r^2 + \overline{B}_\phi^2\right) - \eta\left(\frac{\partial \overline{B}_r}{\partial z}\overline{B}_\phi - \frac{\partial \overline{B}_\phi}{\partial z}\overline{B}_r\right)\right] + \boldsymbol{\mathcal{F}}^{\mathrm{NV}}. \tag{8.3}$$

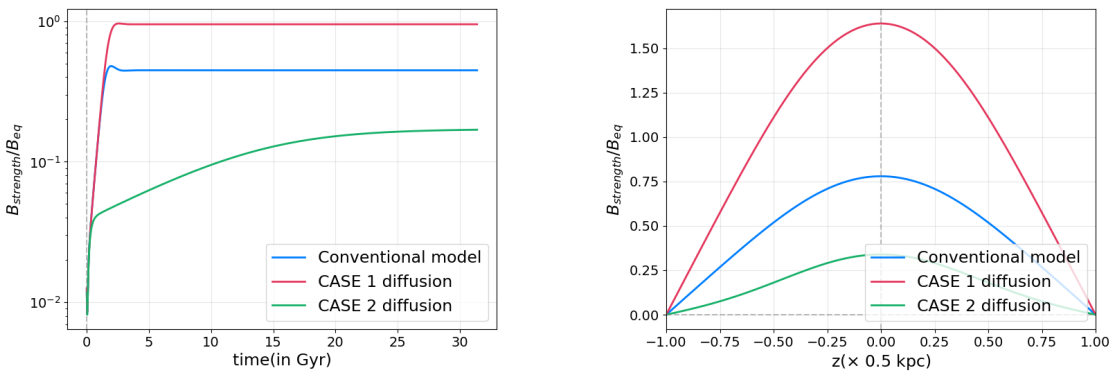
In the early stages, when the mean fields are still weak, the nonlinear quenching and gradient terms are negligible. Thus, Equation 2.8 is not yet active, and α_k along with Ω drive Equations 7.1 and 7.2. Once the mean fields begin to grow, Equation 2.8 starts to influence the system. At this point, the absence of α_k would normally mean no mean field growth. However, since the Vishniac flux can exist even in the absence of mean fields, Equation 2.8 still evolves the system beyond the initial growth phase.

At sufficiently large and positive values, the Vishniac flux term $\boldsymbol{\mathcal{F}}^{\mathrm{NV}}$ directly activates α_m. That is, the helicity transport creates a net gradient in magnetic helicity, mimicking the role of α_k. Therefore, the Vishniac flux-generated α_m results in a non-zero

total α-effect which, together with Ω, forms the positive feedback loop required for classic mean-field dynamo action.

In short, a sufficiently strong Vishniac flux seeds an effective α-effect that is entirely magnetic in origin. This enables the classic dynamo feedback loop without the need for kinetic helicity. Hence, a dynamo is possible that is purely driven by magnetic energy.

8.3 Diffusive Flux Behavior



(a) Mean field strength vs. time. The y-axis is shown in logarithmic scale.

(b) Total field strength vs. z. This is the vector sum of $\overline{B}_\phi$ and $\overline{B}_r$.

Figure 8.4: Comparison of conventional and modified diffusive fluxes for Case 1 and Case 2 velocity profiles. Case 1 and Case 2 correspond to the different velocity profiles discussed in Chapter 3. The blue curve represents the conventional $\alpha\Omega$ model with the standard diffusive flux formulation. The diffusive flux strength is governed by the parameter ξ_0, which is fixed at $\xi_0 = 0.6$. There is no Vishniac flux in these models, as we aim to isolate and analyze the effects of diffusion alone.

The diffusive fluxes derived for Case 1 and Case 2 are examined in Figure 8.4. To better understand the influence of our modified diffusive helicity flux on dynamo growth, we analyze the flux expression in Case 1:

$$-\nabla \cdot \mathcal{F}^D \propto \left[\underbrace{e^{+z^2/h^2}}_{\text{tries to spread outwards}} + \underbrace{\xi_0 e^{-z^2/h^2}}_{\text{tries to contain in midplane}} \right] \frac{\partial^2 h^b}{\partial z^2} \tag{8.4}$$

This structure reveals a clear competition between two opposing effects: the first term tends to drive helicity outward toward the disk boundaries, while the second term, proportional to the parameter ξ_0, attempts to confine helicity within the midplane. We tested this new form of diffusive flux against the standard model considered in [5].

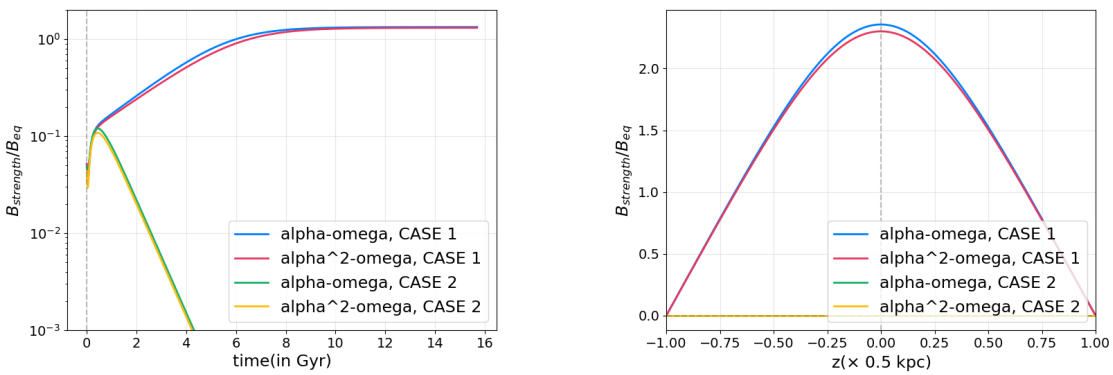
The comparison of magnetic field growth (see Figure 8.4) shows that the growth rates, represented by the slope during the initial exponential phase, are nearly identical for Case 1 and the conventional model. These two models both adopt the same turbulent velocity stratification. The plots shown in Figure 8.4 are taken from the galactic midplane. Since $z = 0$ at this point, the second term in Equation 5.12 vanishes. Thus, the heuristic equation used in both the conventional model and Case 1 diffusion becomes effectively the same. Both are proportional to the second derivative of α_m, differing only in the constant magnitude, which in Case 1 depends monotonically on ξ_0. This explains why the conventional and Case 1 diffusion models have the same growth rate but slightly different, albeit noticeable, saturation strengths.

In Case 2, however, the flux expression lacks any such competition between terms. Here, the entire expression favors confining helicity near the midplane. As a result, diffusion is particularly strong in this region, which suppresses field growth. This is reflected in the lower growth rate observed in the simulations for Case 2. Furthermore, in Case 2, the saturation field strength is significantly lower, likely due to excessive midplane diffusion weakening the dynamo.

For the sake of self-consistency, we will use the new diffusion form given in Equa-

tions 5.12 and 5.33 when including the Vishniac flux in the model, since both of these fluxes are derived using the same technique [9].

8.4 Comparison of the $\alpha\Omega$ vs. $\alpha^2\Omega$ Dynamo Mechanisms



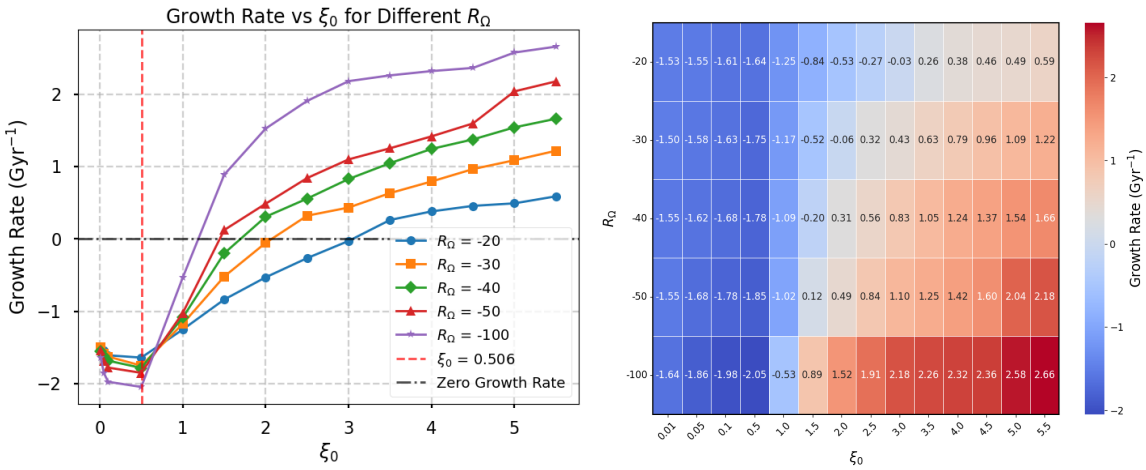
(a) Mean field strength vs. time. The y-axis is in logarithmic scale.

(b) Total field strength vs. z. This is the vector sum of $\overline{B}_\phi$ and $\overline{B}_r$.

Figure 8.5: These models test the impact of new flux terms in the $\alpha\Omega$ versus $\alpha^2\Omega$ dynamo regimes. All models exclude the kinetic α-effect ($\alpha_k = 0$) and use $\xi_0 = 3.5$. Case 1 and Case 2 correspond to the different velocity profiles discussed in Chapter 3.

To examine whether the characteristics of the flux change under a different dynamo regime, we replaced the $\alpha\Omega$ model with an $\alpha^2\Omega$ configuration. The corresponding results are shown in Figure 8.5. We find that the behavior of the flux remains consistent between the two models, with the profiles nearly superimposable. This suggests that the underlying flux dynamics are robust to the choice of dynamo mechanism.

8.5 Parameter Space Study



(a) Here, v_0 is fixed at 20 km/s while R_Ω is varied from -20 to -100.

(b) Contour plot showing how the growth rate varies with ξ_0 and R_Ω.

Figure 8.6: Plots illustrating the dependence of the growth rate on ξ_0 and R_Ω. These models exclude the kinetic α-effect ($\alpha_k = 0$).

We conducted a parameter space study to investigate how the magnetic field growth rate depends on the strength of turbulence and differential rotation. In Figure 8.6a, the turbulent velocity amplitude is fixed at $v_0 = 20$ km/s, while R_Ω is varied from -10 to -80. Our results indicate that changes in v_0 have a minimal effect on the growth rate. However, increasing the magnitude of R_Ω leads to a significant enhancement in the growth rate.

This suggests that for sufficiently large values of $|R_\Omega|$, a dynamo driven purely by the Vishniac flux is feasible even when $\xi_0 < 3.5$.

Chapter 9

An Anaytical Study

9.1 The no-z Approximation

To simplify the one-dimensional dynamo equations given in equations 7.1,7.2 and 7.3 in the vertical direction, we adopt the no-z approximation. This method replaces vertical derivatives with algebraic terms:

$$\frac{\partial}{\partial z} \to \frac{1}{h}, \quad \frac{\partial^2}{\partial z^2} \to -\frac{1}{h^2}$$

Similarly, in no-z approximation for case 1 profile, we have:

$$\frac{\partial b^2}{\partial z} \to \frac{-b^2}{h} \tag{9.1}$$

$$\frac{\partial v^2}{\partial z} \to \frac{v^2}{h} \tag{9.2}$$

and Case 2 becomes:

$$\frac{\partial b^2}{\partial z} \to \frac{-b^2}{h} \tag{9.3}$$

$$\frac{\partial v^2}{\partial z} \to \frac{-v^2}{h} \tag{9.4}$$

The signs are chosen to reflect boundary conditions and h is the disc semi-thickness, the constants used are from [5]. In the steady state, this reduces the system to a set of algebraic equations, effectively representing vertically averaged fields. For the kinematic regime, where magnetic fields grow exponentially like $\mathbf{B}(t) \propto e^{\gamma t}$, the approximation allows us to solve for the growth rate γ directly.

9.2 No-z Approximation on Field Evolution Equations

Under this approximation, the mean field evolution equations become:

$$\gamma B_r = -\frac{1}{h}(\alpha B_\varphi) - \frac{\eta}{h^2} B_r \tag{9.5}$$

$$\gamma B_\varphi = -\Omega B_r - \frac{\eta}{h^2} B_\varphi \tag{9.6}$$

From equation 9.5, solve for B_r and substitute it in equation 9.6 to get:

$$\gamma = \pm\sqrt{\frac{\Omega\alpha}{h}} - \frac{\eta}{h^2} \tag{9.7}$$

Now, we are setting the α_k to 0 to study how the Vishniac flux is alone driving the dynamo, thus $\alpha = \alpha_m$:

$$\gamma = \pm\sqrt{\frac{\Omega\alpha_m}{h}} - \frac{\eta}{h^2} \tag{9.8}$$

The first term is the driving term and the second is the diffusion term. We are interested in growing solutions, so we continue with the positive sign for the first term.

Now, to find the exact value of α_m for which growth starts, we can do the simple case where $\gamma = 0$ and solve for α_m. We get the dynamo condition to be:

$$\alpha_m > \frac{\eta^2}{h^3\Omega}. \tag{9.9}$$

9.3 No-z Approximation for the New Diffusive Flux

Case 1 The expression for the divergence of the diffusive flux in Case 1 is:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^D = \frac{7\tau v_0^2}{27}\left[\exp\left(\frac{z^2}{h^2}\right) + \xi_0 \exp\left(-\frac{z^2}{h^2}\right)\right]\nabla^2 h^b \tag{9.10}$$

$$+ \frac{14 z v_0^2 \tau}{27 h^2}\left[\exp\left(\frac{z^2}{h^2}\right) - \xi_0 \exp\left(-\frac{z^2}{h^2}\right)\right]\partial_z h^b \tag{9.11}$$

At the midplane ($z = 0$), the second term vanishes since it is odd in z, leading to:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^D = \frac{7\tau v_0^2}{27}\left(1 + \xi_0\right)\nabla^2 h^b \tag{9.12}$$

Substituting the expression for the magnetic helicity density proxy,

$$h^b = 12\pi\rho\tau v_0^2 \alpha_m, \tag{9.13}$$

we get:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^D = \frac{7\tau v_0^2}{27}\left(1 + \xi_0\right)\nabla^2\left(12\pi\rho\tau v_0^2 \alpha_m\right) \tag{9.14}$$

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^D = \frac{84\pi\rho\tau^2 v_0^4}{27}\left(1 + \xi_0\right)\nabla^2 \alpha_m \tag{9.15}$$

This form closely resembles standard Fickian diffusion, where the flux is proportional to the Laplacian of the quantity being diffused, in this case α_m. However, here the proportionality constant depends explicitly on physical parameters like τ, v_0, ρ, and the dimensionless factor $(1 + \xi_0)$, providing a physically motivated and stratification-dependent generalization.

Now, applying the no-z approximation (i.e., replacing the second derivative with $-1/h^2$ times the field), we obtain:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^D = -\left(\frac{84\pi\rho\tau^2 v_0^4}{27 h^2}\right)\left(1 + \xi_0\right)\alpha_m \tag{9.16}$$

This expression captures the effective helicity dissipation at the midplane under the assumption that the vertical scale of variation is h.

Case 2 For Case 2, the flux divergence is given by:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^D = \frac{7\tau v_0^2}{27}(1+\xi_0)\exp\left(-\frac{z^2}{h^2}\right)\left[\frac{\partial^2 h^b}{\partial z^2} - \frac{2z}{h^2}\frac{\partial h^b}{\partial z}\right] \tag{9.17}$$

At $z = 0$, the second term inside the bracket vanishes due to its odd z-dependence, leaving:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^D = \frac{7\tau v_0^2}{27}(1+\xi_0)\frac{\partial^2 h^b}{\partial z^2} \tag{9.18}$$

This is structurally identical to the Case 1 expression at $z = 0$, and therefore the no-z approximations for Case 1 and Case 2 yield the same effective diffusive form.

9.4 No-z Approximation for the New Vishniac Flux

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = -(2-q)\Omega\,\hat{z}\cdot\left(C_1\frac{b^2\tau^2}{4\pi\rho}\frac{\partial b^2}{\partial z} + C_2\frac{\partial v^2}{\partial z}\tau^2 b^2 + C_3 v^2\tau^2\frac{\partial b^2}{\partial z} + C_4 v^2\tau^2\frac{\partial b^2}{\partial z}\right)\hat{z} \tag{9.19}$$

Case 1 Using the no-z approximation, the equation simplifies to:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = -(2-q)\Omega\,\hat{z}\cdot\left(C_1\frac{b^2\tau^2}{4\pi\rho}\frac{-b^2}{h} + C_2\frac{v^2}{h}\tau^2 b^2 + C_3 v^2\tau^2\frac{-b^2}{h} + C_4 v^2\tau^2\frac{-b^2}{h}\right)\hat{z} \tag{9.20}$$

At the midplane ($z = 0$) and assuming a flat rotation curve (i.e., $q = 1$), this becomes:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = -\Omega\,\hat{z}\cdot\left[-C_1\cdot\frac{b_0^4\tau^2}{4\pi\rho h} + (C_2 - C_3 - C_4)\cdot\frac{v_0^2\tau^2 b_0^2}{h}\right]\hat{z} \tag{9.21}$$

Factoring out common terms, we have:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = -\Omega \cdot \frac{v_0^2 \tau^2 b_0^2}{h} \left[-C_1 \xi_0 + (C_2 - C_3 - C_4)\right] \tag{9.22}$$

Substituting $b_0^2 = 4\pi\rho v_0^2 \xi_0$ gives:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = -\Omega \cdot \frac{v_0^4 \tau^2 4\pi\rho\xi_0}{h} \left[-C_1 \xi_0 + (C_2 - C_3 - C_4)\right] \tag{9.23}$$

Thus, the final no-z form of the Vishniac flux becomes:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = \Omega \cdot \frac{v_0^4 \tau^2 4\pi\rho\xi_0}{45h} \left(7\xi_0 - \frac{257}{72}\right) \tag{9.24}$$

This expression reveals that the Vishniac flux changes sign depending on the value of ξ_0. The sign determines whether the flux aids or suppresses dynamo action by transporting magnetic helicity out of the system.

We define a critical value of ξ_0, denoted ξ_{crit}, where the flux transitions from suppressing to promoting the dynamo:

$$\xi_{\text{crit}} = \frac{1285}{2520} \approx 0.506 \tag{9.25}$$

Therefore:

- When $\xi_0 > \xi_{\text{crit}}$, the flux becomes helicity-ejecting, thereby alleviating resistive quenching and supporting dynamo action.
- When $\xi_0 < \xi_{\text{crit}}$, the flux suppresses helicity transport, leading to resistive quenching.

Case 2 In this case, the velocity profile is altering the sign of its derivative. The expression for the flux becomes:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = -(2-q)\Omega\,\hat{z} \cdot \left(C_1 \frac{b^2\tau^2}{4\pi\rho}\frac{-b^2}{h} + C_2 \frac{-v^2}{h}\tau^2 b^2 + C_3 v^2\tau^2 \frac{-b^2}{h} + C_4 v^2\tau^2 \frac{-b^2}{h} \right) \hat{z} \tag{9.26}$$

Following similar steps as in Case 1, we obtain:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = -\Omega \cdot \frac{v_0^2\tau^2 b_0^2}{h} \left[-C_1\xi_0 + (-C_2 - C_3 - C_4) \right] \tag{9.27}$$

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} = \Omega \cdot \frac{v_0^2\tau^2 b_0^2}{45h} \left(7\xi_0 - \frac{2503}{360} \right) \tag{9.28}$$

Here, the critical value of ξ_0 for helicity ejection is:

$$\xi_{\text{crit}} = \frac{2503}{2520} \approx 0.993 \tag{9.29}$$

Thus, in this profile:

- For $\xi_0 > 0.993$, the flux promotes helicity ejection and supports dynamo action.
- For $\xi_0 < 0.993$, helicity transport is suppressed, and the dynamo becomes resistively quenched.

The no-z approximation of the new Vishniac flux reveals that the structure of the small-scale field, encapsulated by ξ_0, plays a crucial role in determining whether the flux assists or hinders the dynamo. The transition occurs sharply at ξ_{crit}, which is sensitive to the vertical profile of the turbulence and field components.

9.5 No-z Approximation in the Helicity Evolution Equation

During the initial kinematic regime, the generated mean magnetic fields are weak, and their contribution to the evolution of magnetic helicity is negligible. Thus, the first term in Eq. (7.3) drops out, leaving only the divergence of the Vishniac and diffusive helicity fluxes:

$$\frac{\partial \alpha_m}{\partial t} = -\nabla \cdot \boldsymbol{\mathcal{F}^D} - \nabla \cdot \boldsymbol{\mathcal{F}^{NV}}. \tag{9.30}$$

Clearly, there is no growth of the magnetic component of the α-effect, α_m, unless there is also a kinetic contribution α_k. In the absence of both, large-scale magnetic field growth is suppressed. Let us now evaluate these helicity flux contributions using the no-z approximation.

Case 1

$$\frac{\partial \alpha_m}{\partial t} = -\left(\frac{84\pi\rho\tau^2 v_0^4}{27h^2}\right)(1+\xi_0)\alpha_m + \Omega\frac{v_0^4\tau^2 4\pi\rho\xi_0}{45h}\left(7\xi_0 - \frac{257}{72}\right) \tag{9.31}$$

Assuming a steady state ($\partial_t \alpha_m = 0$):

$$\alpha_m = \frac{\Omega h \xi_0}{2520(1+\xi_0)}(504\xi_0 - 257) \tag{9.32}$$

Substituting into the dynamo condition $\alpha_m > \eta^2/(h^3\Omega)$, we obtain:

$$\frac{\Omega h \xi_0}{2520(1+\xi_0)}(504\xi_0 - 257) > \frac{\eta^2}{h^3\Omega} \tag{9.33}$$

$$\Rightarrow \frac{\xi_0}{2520(1+\xi_0)}(504\xi_0 - 257) > \frac{1}{R_\Omega^2}, \tag{9.34}$$

where $R_\Omega = \Omega h^2/\eta$ is the shear-based magnetic Reynolds number.

For the fiducial model with $R_\Omega = -20$, we obtain:

$$504\xi_0^2 - 263.3\xi_0 - 6.3 > 0. \tag{9.35}$$

Solving the quadratic yields:

$$\xi_0 < -0.0229 \quad \text{or} \quad \xi_0 > 0.5453. \tag{9.36}$$

We define another critical value of ξ_0, denoted ξ_{growth}, where the Vishniac flux alone can lead the dynamo action. Since ξ_0 is a ratio of energies, the only physically meaningful solution is:

$$\xi_{\text{growth}} = 0.5453 \tag{9.37}$$

Case 2

$$\frac{\partial \alpha_m}{\partial t} = -\left(\frac{84\pi\rho\tau^2 v_0^4}{27h^2}\right)(1+\xi_0)\alpha_m - \Omega\frac{v_0^2\tau^2 b_0^2}{45h}\left(7\xi_0 - \frac{2503}{360}\right) \tag{9.38}$$

In steady state, solving for α_m:

$$\alpha_m = -\frac{\Omega h\xi_0(2520\xi_0 - 2503)}{12600(1+\xi_0)} \tag{9.39}$$

Substituting into the dynamo condition:

$$-\frac{\Omega h\xi_0(2520\xi_0 - 2503)}{12600(1+\xi_0)} > \frac{\eta^2}{h^3\Omega} \tag{9.40}$$

leads to the inequality:

$$5040\xi_0^2 - 4943\xi_0 + 63 < 0 \tag{9.41}$$

Solving this quadratic gives the critical value for growth as:

$$\xi_{\text{growth}} = 0.9678 \tag{9.42}$$

9.6 Interpretation of No-z Analysis

In both cases, the critical value ξ_{crit} marks a threshold beyond which the helicity fluxes are sufficient to support dynamo action. The physical insight is that a sufficiently strong small-scale magnetic energy (relative to kinetic energy) leads to enhanced magnetic helicity fluxes. These fluxes help to offset resistive losses and enable sustained growth of large-scale magnetic fields.

Thus, whenever the value of ξ_0 in the system exceeds ξ_{crit}, the Vishniac flux contributes positively to the dynamo mechanism. Moreover, its effectiveness increases with increasing ξ_0. This behavior explains the trends observed in Figures 8.1 and 8.2 in Section 8.1.

The critical value obtained for Case 1 matches well with simulation results. For Case 2, while the match is not exact, the qualitative trend remains consistent. The discrepancy can likely be attributed to midplane-dominated diffusion effects, as discussed in Section 8.3. Nevertheless, the overall behavior is as predicted by the no-z approximation.

From the no-z analysis applied to the dynamical quenching model—specifically the dynamo condition and the inequality involving α_m, we found that dynamo growth requires α_m to exceed a certain positive threshold, which depends on turbulent diffusivity and the disc scale height. This insight helps explain why dynamo action can occur due to the Vishniac flux alone, even in the absence of a kinetic α-effect α_k.

At large ξ_0, i.e., in the regime of magnetically dominated turbulence, the Vishniac flux becomes dominant. Its positive divergence drives an increase in α_m, as seen from the

helicity evolution equation (Equation 9.30). This means α_m contributes significantly to the total α-effect from the onset of dynamo action, unlike in conventional dynamos where mean fields must grow first to accumulate small-scale helicity and generate α_m, as discussed in Section 2.5.

Here, α_m effectively plays the role of α_k in traditional dynamos. It triggers the positive feedback loop required for the dynamo instability, although the origins and physical interpretations of the two alpha-effects are distinct.

From the no-z approximation, we found the critical values of ξ_0 required for Vishniac-flux-driven dynamo action to be:

- Case 1: $\xi_{\text{growth}} = 0.5453$
- Case 2: $\xi_{\text{growth}} = 0.9678$

In simulations, the corresponding thresholds were found to be approximately 3.5 and 6.5 for Case 1 and Case 2 respectively. This discrepancy is expected, as the no-z approximation is a simplified model intended to provide qualitative understanding rather than precise numerical predictions. The key result is that the no-z analysis correctly predicts that Vishniac flux alone can sustain dynamo growth beyond a critical ξ_0, and that this critical value is significantly higher in Case 2 than in Case 1, by approximately a factor of two, which is consistent with the simulation results.

Chapter 10

Conclusion

Conventional mean-field dynamo theory struggles to fully explain the large-scale magnetic fields observed in galactic interstellar media (ISMs). The models typically relied upon to account for large-scale magnetic fields remain incomplete and require further development. This study aimed to address some of these gaps by incorporating a previously unexplored magnetic helicity flux, the Vishniac flux, into a dynamical quenching mean-field model. A central objective of this thesis was to determine whether the Vishniac flux acts as an inhibitor or supporter of conventional dynamo action, and whether this flux alone, in the absence of the conventional drivers of dynamo theory, can give rise to dynamo action.

This project was based on the analytical work by Gopalakrishnan & Subramanian (2023), where they derived a flux—originally hypothesized by Vishniac in 2012—that could potentially take over dynamo action and shift our understanding of how the dynamo operates. This flux was obtained from third-order correlation terms of small-scale fields. In this thesis, both the Vishniac flux and a new form of diffusive flux were studied. The diffusive flux was mathematically derived for the first time in their work; prior to this, it had been modeled heuristically using a Fickian profile. This thesis marks the first time both the Vishniac flux and the newly formulated diffusive flux have been incorporated into a simulation and studied in a dynamical quenching model.

A major component of this thesis involved code development, which was carried out independently in `FORTRAN` using a sixth-order finite differencing scheme and an implicit Runge–Kutta method of third order (RK3) for time stepping. The code was rigorously tested by reproducing plots from published works. Once validated, the new terms—namely, the Vishniac flux and the diffusive flux—were implemented to simulate the model and study the dynamo action.

We found that the Vishniac flux alone can, in principle, drive dynamo action. When examining very high values of ξ_0, the flux becomes significantly stronger due to its quadratic dependence on ξ_0. Under these conditions, the Vishniac flux by itself can sustain dynamo action. While this scenario is not realistically applicable to typical galactic systems, it is theoretically intriguing. It represents a form of magnetically driven dynamo action—a departure from the conventional kinetically driven models—adding a new layer of complexity to our understanding of dynamo processes.

Our primary question was whether this new flux could significantly alter our understanding of mean-field dynamo theory. The short answer is: theoretically, yes; realistically, not by much. There exists a threshold value of ξ_0, denoted ξ_{growth}, above which independent dynamo action becomes possible. However, in all our simulations, this threshold occurs at superequipartition magnetic field strengths. While not unphysical, such conditions are not realistically expected in galactic environments. Thus, we conclude that the Vishniac flux alone does not independently drive a dynamo to saturation for realistic values of ξ_0.

Next, we explored whether adding this flux to a conventional dynamo, where α_k drives the system, would enhance or hinder magnetic field growth. The answer depends on the circumstances. The flux expression introduces a saddle point where it changes

sign, and this transition occurs at different values of ξ_0 depending on the turbulence profile. For turbulence peaking at the disc edges, the sign flip occurs around $\xi_0 = 0.506$, whereas for turbulence peaking at the midplane, it occurs closer to $\xi_0 = 1$. Below these values, the Vishniac flux suppresses dynamo action, while above them, it enhances it.

We also performed a parameter space study by varying the rotation parameter R_Ω, which is directly proportional to the strength of the Vishniac flux. As R_Ω increases, the critical threshold ξ_{growth} decreases, making independent dynamo action by Vishniac flux more possible at lower values of ξ_0.

We analyzed the simulation results using a method called the no-z approximation. The findings from the simulations—namely, the independent dynamo action and the regimes of dynamo support and inhibition by the Vishniac flux—were successfully validated. This qualitative analysis helped us understand how the Vishniac flux alone can drive dynamo action without the conventional α-effects. Specifically, at sufficiently high ξ_0, the Vishniac flux produces a non-zero value of α_m even during the initial phase of dynamo action. We found that this α_m mimics the α_k effect by forming a positive feedback loop with the Ω effect, thereby driving the dynamo instability.

One key realization from this thesis is that in systems with either very strong small-scale magnetic field gradients or very large values of rotation, the Vishniac flux can become a significant effect. In such systems, it can drive a mean-field dynamo in the model and generate large-scale magnetic fields without kinetic effects. This implies the existence of a purely magnetically driven dynamo, which is a very exciting prospect for future research. Further work could explore such systems in more detail

to study this magnetically driven dynamo action.

A major limitation of the model studied here is the assumption of a vertically constant density profile. This is unrealistic, although we did investigate different turbulent profiles and found that the underlying effect of the Vishniac flux remained qualitatively the same. This is an area where the model can be improved.

In this study, we considered two profiles for ξ: one that decreases with z and another that remains constant. A natural next step is to investigate the system with a ξ profile that increases with z. An initial attempt was made in this direction, but during implementation, we encountered an issue where α_m attained large unphysical values near the boundary, leading to numerical instability. This should be looked into further and would be a good next step for this project.

Another obvious direction for improvement is to relax the thin-disc approximation. Incorporating the galactic corona and solving the system in two dimensions could yield global dynamo solutions and potentially offer deeper insights into the underlying physics.

In summary, this thesis has advanced our understanding of the galactic dynamo theory by incorporating and rigorously analyzing the Vishniac flux within a dynamical quenching framework. The finding of a potentially purely magnetically driven dynamo opens exciting new avenues for theoretical and observational investigation. Future studies building on this work, especially with more realistic galactic models and multidimensional simulations, promise to deepen our insight into the complex mechanisms that sustain cosmic magnetic fields.

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